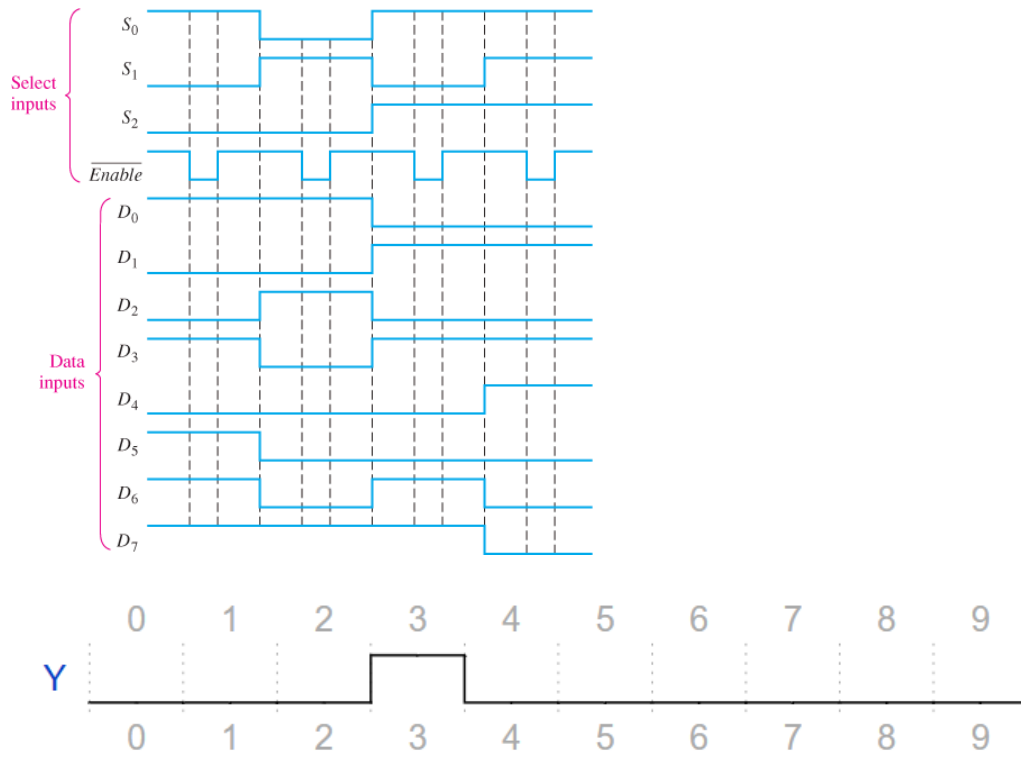
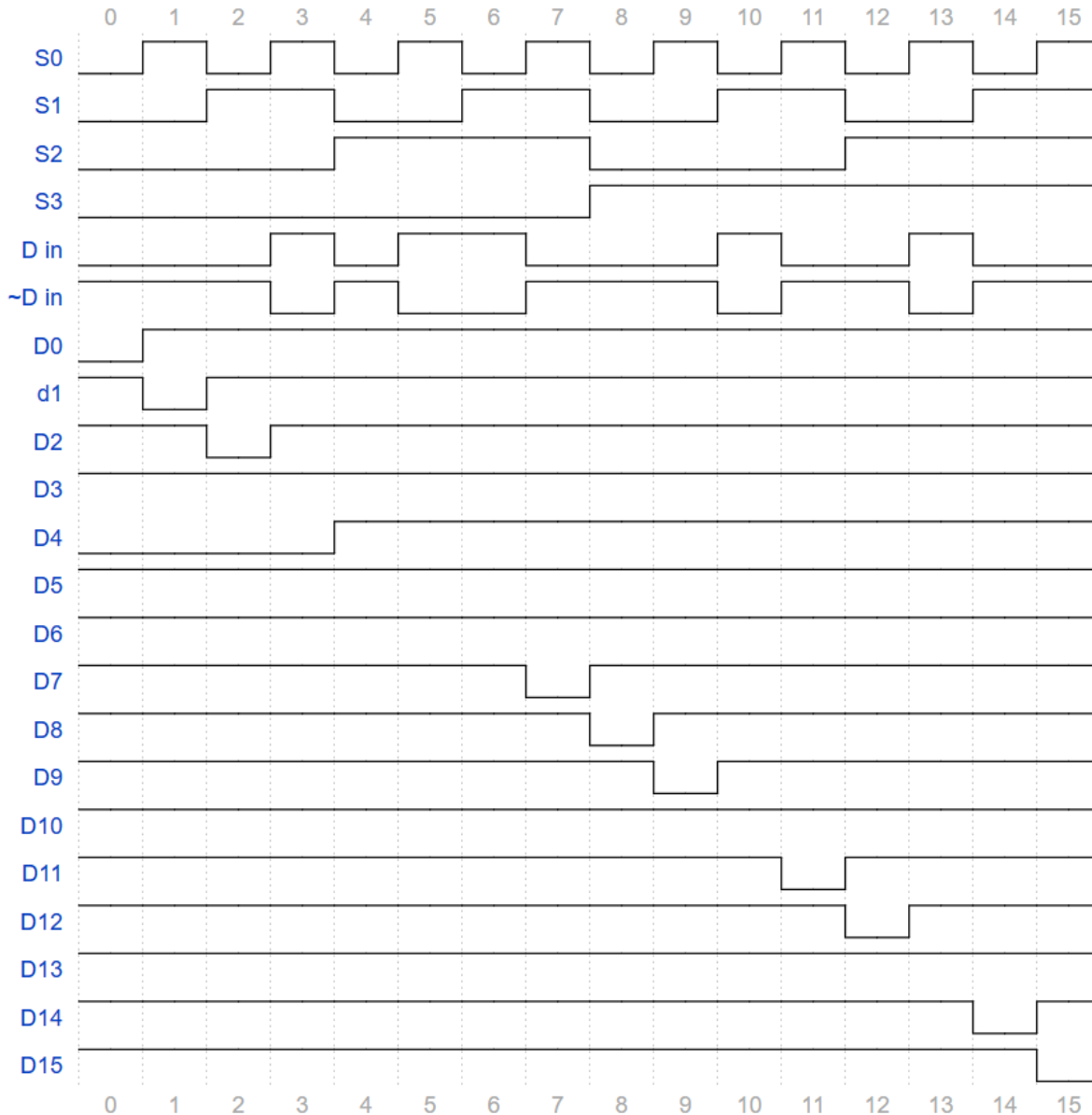


30) The waveforms in Figure 6–81 are observed on the inputs of a 74HC151 8-input multiplexer. Sketch the Y output waveform.

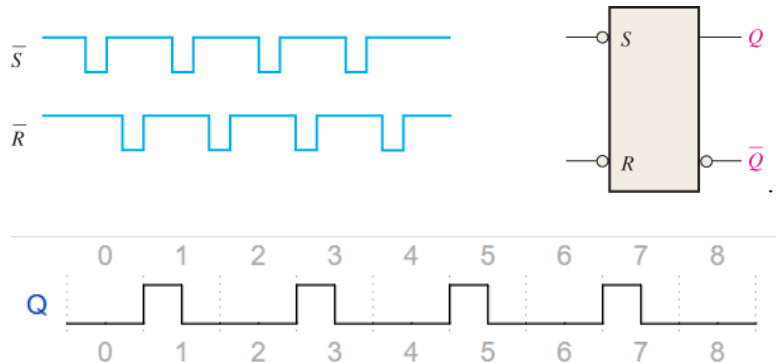


31) Develop the total timing diagram (inputs and outputs) for a 74HC154 used in a demultiplexing application in which the inputs are as follows: The data-select inputs are repetitively sequenced through a straight binary count beginning with 0000, and the data input is a serial data stream carrying BCD data representing the decimal number 2468. The least significant digit (8) is first in the sequence, with its LSB first, and it should appear in the first 4-bit positions of the output.

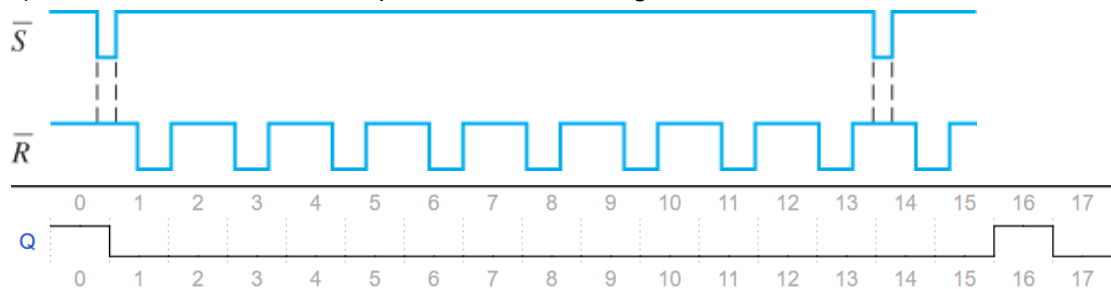


CH 7

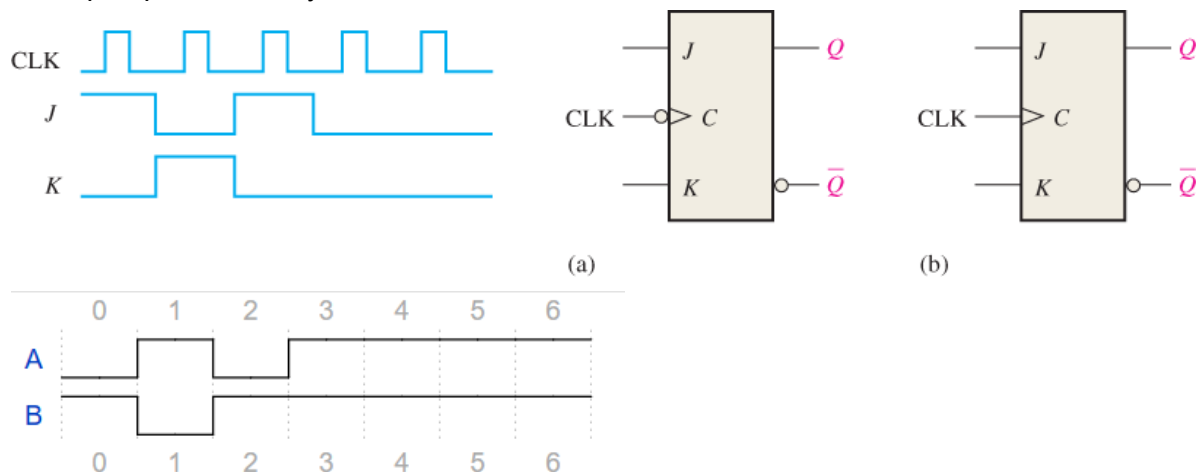
1) If the waveforms in Figure 7–70 are applied to an active-LOW input $\overline{S}$ latch, draw the resulting Q output waveform in relation to the inputs. Assume that Q starts LOW.



3) Solve Problem 1 for the input waveforms in Figure 7–72.

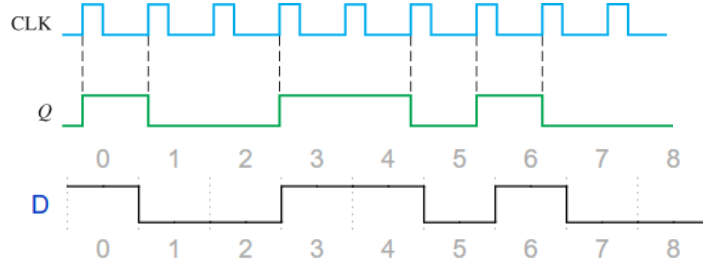


8) Two edge-triggered J-K flip-flops are shown in Figure 7–77. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET.

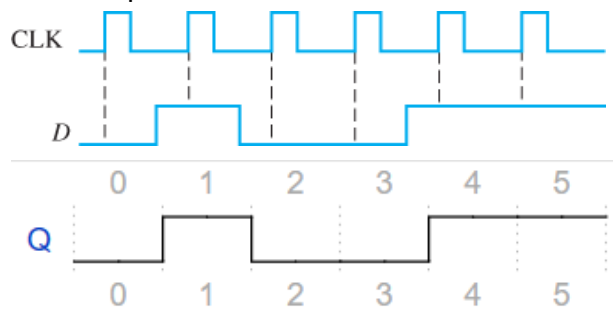


Both (a) and (b) have the same number of clock pulses, however, the triggering is reversed.

9) The Q output of an edge-triggered D flip-flop is shown in relation to the clock signal in Figure 7–78. Determine the input waveform on the D input that is required to produce this output if the flip-flop is a positive edge-triggered type.



12) For a positive edge-triggered D flip-flop with the input as shown in Figure 7–81, determine the Q output relative to the clock. Assume that Q starts LOW.



21) The data sheet of a certain flip-flop specifies that the minimum HIGH time for the clock pulse is 30 ns and the minimum LOW time is 37 ns. What is the maximum operating frequency?

$$30 + 37 = 67$$

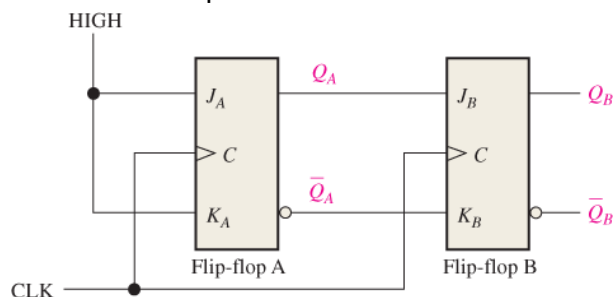
$$1/67 = 0.015 = 15\text{MHz}$$

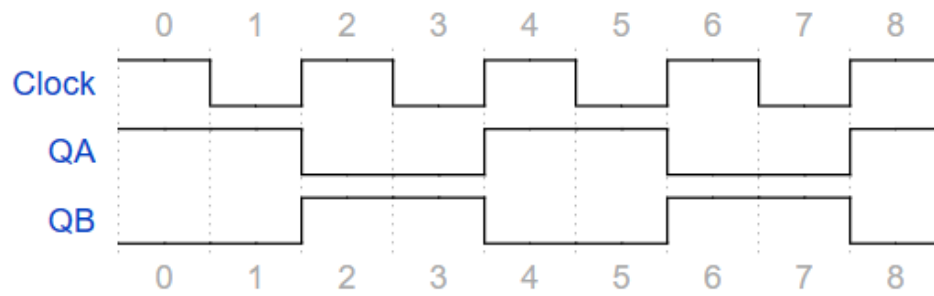
23) The direct current required by a particular flip-flop that operates on a +5V dc source is found to be 10 mA. A certain digital device uses 15 of these flip-flops. Determine the current capacity required for the +5V dc supply and the total power dissipation of the system.

$$15 \times 10\text{mA} = 150\text{mA}$$

$$5\text{V} \times 150\text{mA} = 750\text{mW}$$

26) For the circuit in Figure 7–89, develop a timing diagram for eight clock pulses, showing the QA and QB outputs in relation to the clock.





CH 8

7) What is the state of the register in Figure 8–49 after each clock pulse if it starts in the 101001111000 state?

